

EEET202

CHINES & TRANSFORMERS

MODULE:3

re:6 – Testing of DC Motors

of operation of DC motors and select appropriate motor types for

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Syllabus

generation of torque, torque equation, performance characteristics – starters – 3 point and 4 point starters (principle only).

tors - field control, armature control. Braking of dc motors. Power and efficiency. Testing of dc motors - Swinburne's test, Hopkinson's test. DC motor applications – numerical problems.

Swinburn's Test (No Load Test)

Method

Parameters measured separately

Armature resistance, field resistance, no-load current, no-load power, efficiency at any desired load can be determined.

Applicable to constant flux machines.

Swinburne's Test (No Load Test)

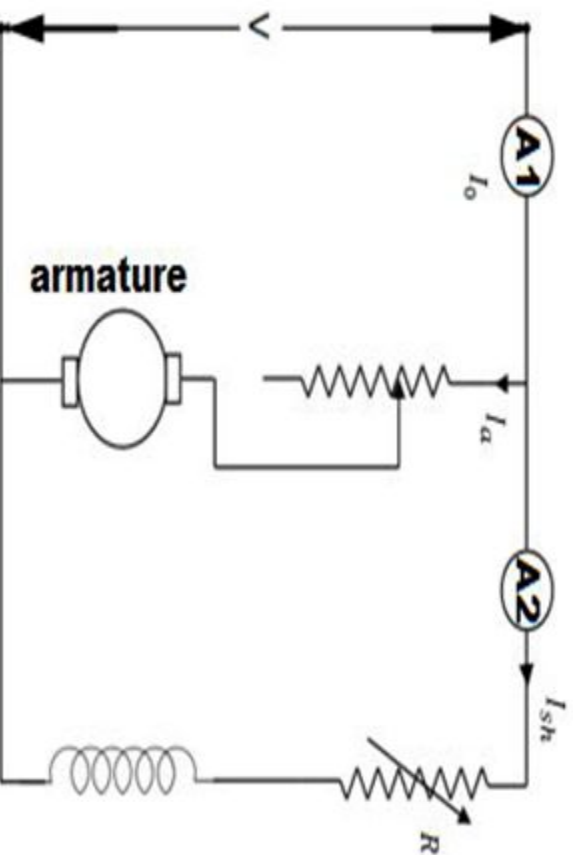
applied across the motor at no load condition.

ted to the rated speed by adjusting the field

rent drawn by the motor (I_o) is measured by the

nt (I_{sh}) is given by the ammeter A2.

the current will be:



Swinburne's Test (No Load Test)

Power drawn from the DC supply during no-load supplies

Core losses / Magnetic losses

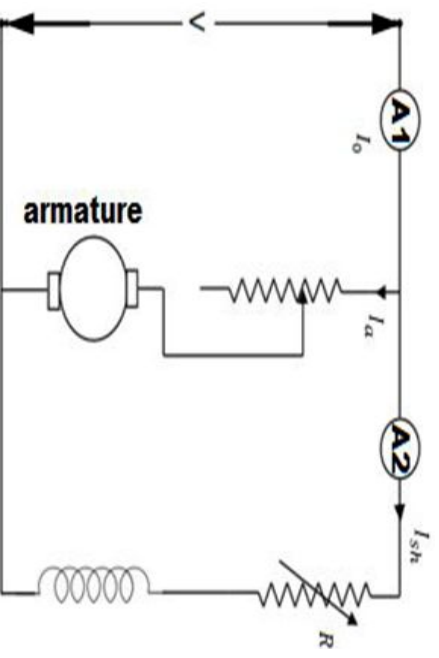
Armature

losses

Field winding

Armature losses vary with load conditions; whereas all other losses remain constant (independent of load conditions)

$$W_c = VI_o - (I_o - I_{sh})^2 R_a$$



Swinburne's Test (No Load Test)

condition (Load current = I_L), Efficiency of DC Machine determined as follows:

(got from swinburne's test)

Generating Mode

$I_L = I_a + I_{sh}$	✓ Armature current, $I_a = I_L + I_{sh}$
$P_{out} = V I_L$	✓ Output Power $P_{out} = V I_L$
$W_c = I_a^2 R_a$	✓ Armature copper loss = $I_a^2 R_a$
$W_a = I_a^2 R_a$	✓ Total losses = $W_c + I_a^2 R_a$
Losses / Input	✓ Efficiency = Output / (Output + Losses)

Turner's Test (No Load Test)

Disadvantages

nt and
e power
a large
That is,
ower.
y load
be

- ✓ Possibility of change in iron losses at various load conditions are not considered.
- ✓ As the test is at no-load, it is impossible to know whether commutation would be satisfactory at full-load and whether the temperature rise would be within the specified limit

on's Test (Back to Back Test)

dc machines are mechanically coupled.

uns as generator and other as a motor.

output of motor drives the generator and the
out of generator is supplying the electrical input

no losses in the machines, they would have run
external power supply.

the losses associated with the machines,
out of generator will not be sufficient to run the
e versa.

re supplied by the external DC source.

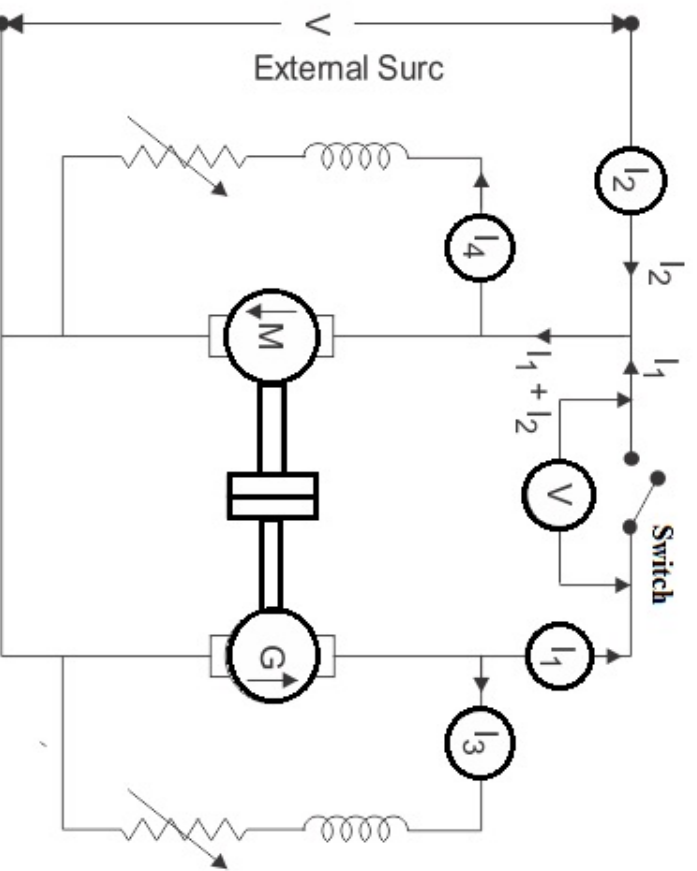
on's Test (Back to Back Test)

started as motor at rated voltage from the supply
which is kept open.

ed to rated speed.

es the Machine G as generator and voltage is read

resistance is
make the
ing as zero.
meter reads
be closed.



on's Test (Back to Back Test)

as and efficiency

om supply = VI_2 = Total losses associated with both

r loss in generator = $(I_1 + I_3)^2 R_a$

r loss in motor = $(I_1 + I_2 - I_4)^2 R_a$

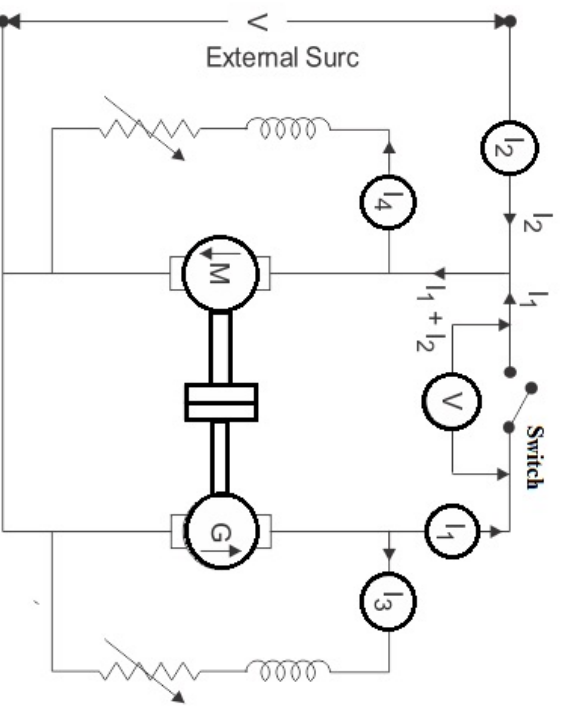
ss in generator = VI_3

ss in motor = VI_4

associated with both

$W_c = VI_2$ - Total Copper

per machine = $W_c / 2$



Don's Test (Back to Back Test)

Disadvantages

- ✓ the test is small
- ✓ tested under full-load
- ✓ temperature rise and the
- ✓ efficiencies can be observed.
- ✓ Two identical machines are required.

Retardation Test

can be determined by conducting retardation
ing the copper losses at any load conditions,
e predetermined.

unning as motor at its rated speed is cut off
, by keeping the field excited.

rmature slows down and its kinetic energy is
e rotational losses (constant losses).

$$= \frac{1}{2} (I_a^2)$$

W = rate of loss of KE

dt (KE)

$(d/dt (w))$

